

THE MYSTERY OF FIRE

What does a flame need to keep burning?



AIR IS EVERYWHERE

You cannot see it — but it can affect fire.



WHAT IS BURNING?

Heat + Fuel + Oxygen → Combustion → Heat + Light





LET'S TEST IT

Candle · Glass jar · Lid · Clay · Board · Lighter (teacher only)

THE EXPERIMENT

A candle. A transparent jar. Controlled openings.



WHAT WILL HAPPEN?

A strong · B weaker · C out



A — KEEPS BURNING



B — BECOMES WEAKER



C — GOES OUT



THE AIR CAN'T BE REPLACED

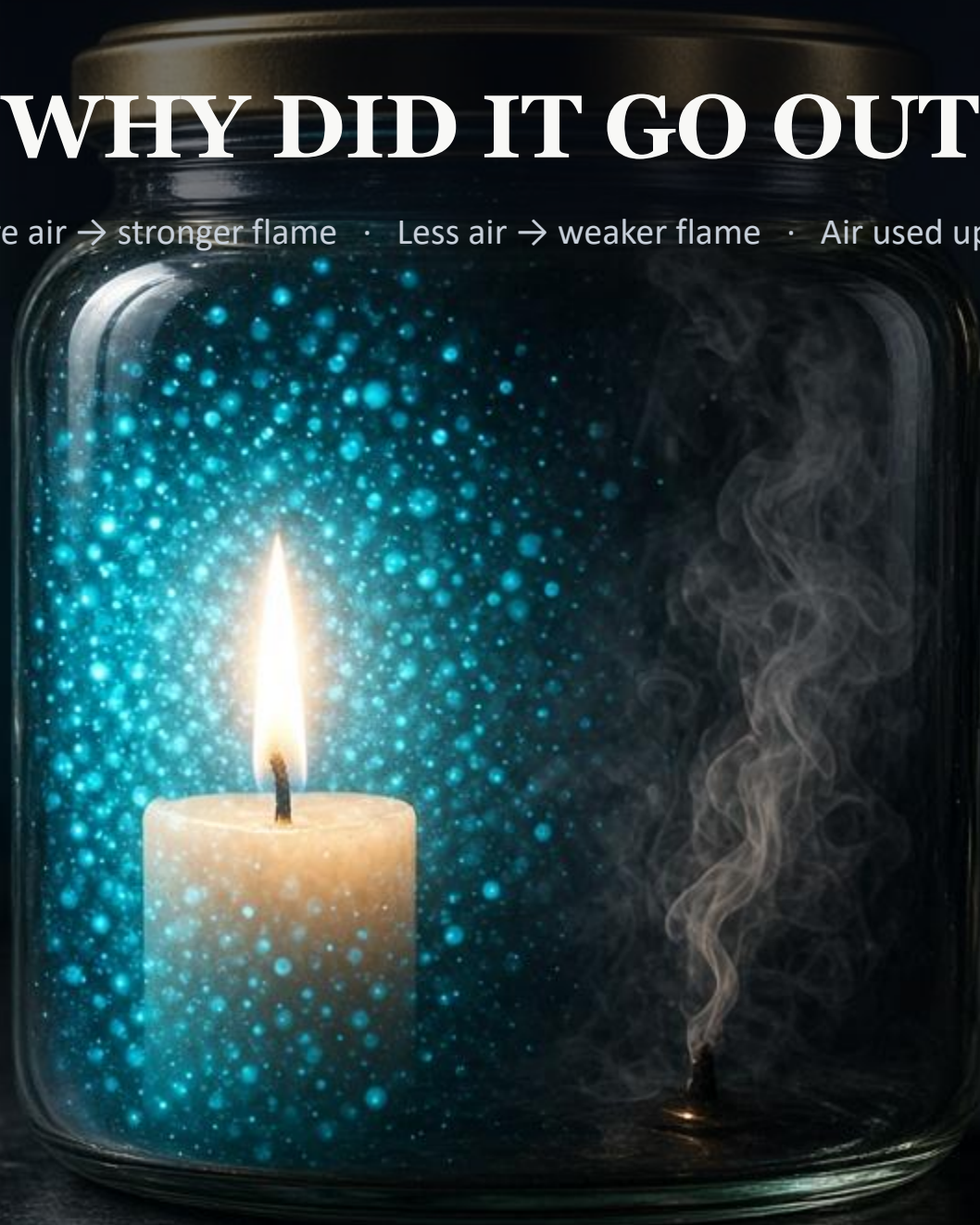
WATCH CAREFULLY

Strong → Weaker → Out



WHY DID IT GO OUT?

More air → stronger flame · Less air → weaker flame · Air used up → out



WHAT IF FRESH AIR CAN ENTER?

A path for new air — but renewal still matters.



FRESH AIR ENTERS

A diagram illustrating the process of convection in a closed system. A lit candle sits inside a clear glass jar, which is placed on a wooden base. The jar is supported by two small glass feet. The candle's flame is bright yellow and orange. From the flame, several thin, wavy lines of orange and yellow light rise towards the top of the jar, representing warm air. At the base of the jar, several thin, wavy lines of blue light curve from the sides towards the center, representing fresh air entering. The background is a dark, solid color.

WARM AIR ↑

OXYGEN → FLAME

FRESH AIR ↓



IT KEEPS BURNING

Fresh air keeps supplying what the flame needs.

CLOSED vs OPEN



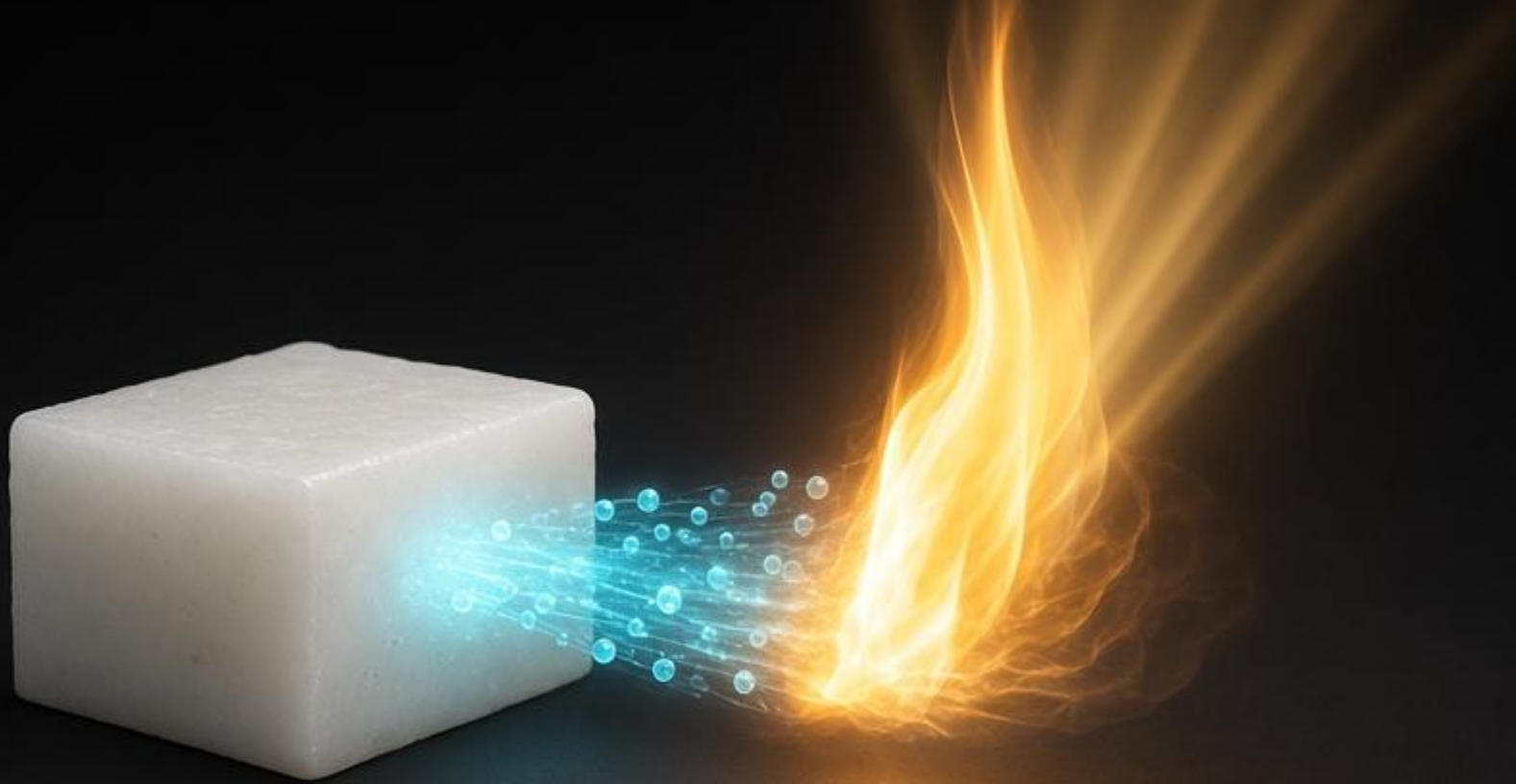
AIR RUNS OUT



AIR RENEWED

WHY DOES THIS HAPPEN?

FUEL + OXYGEN + HEAT → COMBUSTION → HEAT + LIGHT





AIR PLAYS A KEY ROLE IN BURNING

REAL LIFE

Fireplace · Candle · Stove · Engine — each needs air.



WHICH ONE WILL KEEP BURNING?

Enclosed · vs · Airflow



REMEMBER



BURNING NEEDS OXYGEN

FRESH AIR MUST BE SUPPLIED

WITHOUT RENEWAL, THE FLAME GOES OUT